# THANGASAMY GANESAN

**Edge AI & MLOps Expert | AWS Certified Machine Learning - Specialty**

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## PROFESSIONAL SUMMARY

Edge AI & MLOps Expert with 13+ years of experience in Artificial Intelligence (AI), Machine Learning Operations (MLOps), Generative AI, DevOps, AWS, Azure and Linux system administration. Currently leading a 15+ member team and serving as technical advisor for on-device AI on Dell AI PCs and laptops, delivering Small Language Models (SLMs), computer vision, audio and behavioral-biometric models optimized for Neural Processing Unit (NPU) inference under real system-performance constraints. Skilled in agentic AI engineering with Model Context Protocol (MCP) servers and Spec-Driven Development (SDD). Proven record across cloud and edge ML pipelines, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG) and Kubernetes CI/CD, delivering 40% faster model deployment and 25% higher response accuracy.

## TECHNICAL SKILLS

**Edge AI:** On-device inference, NPU acceleration, ONNX Runtime, OpenVINO, MediaPipe, OpenCV, FP16/INT8/INT4 quantization
**Small Language Models (SLMs):** Microsoft Phi-3, Llama 3.2, Gemma, ONNX Runtime GenAI, LoRA fine-tuning, on-device RAG
**AI PC Sensors:** RGB/IR camera, microphone, keyboard/touchpad, presence/ToF, ambient light sensor, accelerometer/hinge-angle
**System Performance:** CPU/GPU/NPU load balancing, thermal & battery-aware inference, sleep/resume, memory/latency profiling
**Computer Vision & Biometrics:** Face verification, pose estimation, rPPG, liveness/anti-spoofing, keystroke dynamics, YOLOv5, BRISQUE
**Generative AI & LLMs:** GPT-4/GPT-3.5, LLMs, RAG, LangChain, LangGraph, LangSmith, Hugging Face, BERT, CLIP, OpenSearch
**Agentic AI & SDD:** MCP servers (Jira, Confluence, GitHub), AI coding agents (Devin, Windsurf), Spec-Driven Development (SDD)
**MLOps & Edge MLOps:** MLflow, Airflow, DagsHub, SageMaker, Azure ML Studio, Azure Databricks, model versioning, drift monitoring
**ML & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, Linear & Logistic Regression, KNN, Decision Trees, Boosting, BM25, fastText
**Cloud - AWS:** EC2, S3, RDS, VPC, IAM, Lambda, ECS, Route 53, CloudFormation, CloudWatch, CodePipeline, SageMaker
**Cloud - Azure:** Azure DevOps, AKS, Data Lake, Data Factory, SQL Database, Cosmos DB, Functions, Blob Storage, Service Bus, Azure AD
**DevOps & CI/CD:** Docker, Kubernetes, Terraform, Ansible, GitHub Actions, Jenkins, GitLab CI/CD, JFrog, SonarQube, Coverity, Jira
**Programming:** Python (NumPy, Pandas, Matplotlib, Seaborn, PyQt6, pytest), Bash/Shell scripting, PowerShell, Jupyter
**Monitoring & Visualization:** Prometheus, Grafana, ELK Stack, New Relic, Azure Monitor, CloudWatch, Nagios, Tableau
**Linux & Virtualization:** RHEL, CentOS, Ubuntu, Debian, VMware, KVM, VirtualBox
**Leadership:** Team leadership (15+ members), technical advisory, mentoring, architecture reviews, stakeholder management

## PROFESSIONAL EXPERIENCE

**Edge AI & MLOps Expert | Dell Technologies, Bengaluru** **Aug 2025 - Present**

*Project: On-Device AI Health & Security System for Dell AI PCs and Laptops*

- Led and mentored a 15+ member cross-functional engineering team and acted as technical advisor on AI PC architecture, model optimization, sensor strategy, architecture reviews and privacy-by-design.
- Architected an on-device Python/PyQt6 AI system using native laptop sensors for wellness insights and presence-based security, with 100% local, in-memory processing and no biometric data persistence or cloud transmission.
- Mapped native and relevant sensors to each test scenario: RGB/IR camera (rPPG, face verification, liveness), microphone (voice stress), keyboard/touchpad (typing biometrics), human-presence/Time-of-Flight (ToF) sensor (walk-away lock, wake-on-approach), ambient light sensor (low-light compensation) and accelerometer/hinge-angle (lid state).
- Developed health models: remote photoplethysmography (rPPG) heart-rate trends, fatigue detection (blink, yawn, typing biometrics), stress estimation (facial action units, Librosa voice pitch) and posture monitoring (MediaPipe pose estimation, OpenCV).
- Built presence-aware security: multi-signal intelligent auto-lock, pre-login face verification, keystroke-dynamics continuous authentication and liveness detection/anti-spoofing using micro-movement and challenge-response.
- Deployed Small Language Models (SLMs: Phi-3, Llama 3.2, Gemma) on-device via ONNX Runtime GenAI with INT4 quantization and LoRA fine-tuning for offline health summaries and a private assistant with on-device RAG.
- Optimized NPU inference with ONNX Runtime and OpenVINO (FP16/INT8, adaptive frame skipping) and handled system-performance scenarios: CPU/GPU/NPU load balancing, thermal-throttling and battery-aware inference, sleep/resume and lid-close handling, and memory/latency profiling, keeping CPU utilization under 15%.
- Established an Edge MLOps workflow: model conversion (PyTorch/TensorFlow to ONNX), quantization, on-device latency/accuracy benchmarking, model versioning and automated CI testing (pytest, mypy).
- Integrated MCP servers (Jira, Confluence, GitHub) with AI coding agents (Devin, Windsurf) and ran Spec-Driven Development (SDD) agent skills end to end: spec/story generation, implementation, AI code review, test generation, CVE and secret-leak scanning.

**MLOps Expert | Schneider Electric, Bengaluru** **May 2023 - Oct 2024**

- Managed a team of data engineers and developers creating AI/ML applications that increased operational efficiency by 25%.
- Developed and deployed a Retrieval-Augmented Generation (RAG) pipeline with LLMs and OpenSearch (Vector DB), integrating external knowledge sources to improve the accuracy and relevance of AI-generated content.
- Implemented RAG strategies with LangChain and LangGraph to optimize LLM performance in real-time applications, increasing response accuracy by 25% in customer-support scenarios.
- Integrated LLMs into existing knowledge management systems to enable contextually accurate Generative AI solutions.
- Architected scalable ML pipelines using Kubernetes and Docker, reducing model deployment time by 40%.

***Key Projects***

- **Conversational Site Search:** Gen AI (LLM) and NLP-powered site search that understands and responds to customer queries conversationally.
- **Competitor Products Intel:** AI-driven gathering, analysis and insight generation from competitors' product data.
- **Single Line Diagram:** AI automation of electrical single-line diagram (SLD) generation; Linear Regression model development and deployment.

***Tech Stack:*** *OpenAI GPT-4/GPT-3.5, LangSmith, Hugging Face Transformers, BERT, CLIP, PyTorch, TensorFlow, YOLOv5, BRISQUE, BM25, fastText, KNN, MLflow, Airflow, DagsHub, AWS SageMaker, Azure ML Studio, AKS, GitHub Actions, Terraform*

**MLOps Engineer | Wells Fargo, Bengaluru** **Aug 2020 - May 2023**

- Designed and implemented MLOps pipelines for credit risk assessment, fraud detection, clinical healthcare analytics and customer churn prediction.
- Built and deployed customer segmentation and churn models with Azure Machine Learning Studio; used Azure Databricks for large-scale data processing and feature engineering.
- Integrated ML pipelines with Azure DevOps CI/CD, monitored production models for performance and drift, and partnered with data scientists, quants and business stakeholders to operationalize and communicate ML value.

***Key Projects***

- **Smart Consumer 360 Predictor:** Predicted online-shopping consumer behavior using data mining, EDA, decision trees, feature correlation and boosting classifiers; achieved 87% accuracy.
- **EDA & Visualization (Banking & Finance):** Qualitative and quantitative analysis of high-volume databases to identify trends; built Python/Jupyter dashboards and reports for business decision-making.
- **Business Analysis / Price Prediction:** Survey data extraction, cleaning and normalization in Azure Machine Learning Studio; Linear Regression model development, deployment and evaluation; led a team of data analysts.

***Tech Stack:*** *Azure Data Lake, Azure Data Factory, Azure SQL Database, AKS, Docker, Kubernetes, Jenkins, GitLab, SonarQube, JFrog, Jira, New Relic, MLflow, Airflow, Scikit-learn, Logistic Regression, Pandas, NumPy, Matplotlib, Seaborn, Tableau*

**Senior DevOps Engineer | L&T Technology Services, Bengaluru** **Aug 2018 - Jun 2020**

*Build management for FCA and Ford Advanced Driver Assistance Systems (ADAS) projects - Aptiv (Poland & India)*

- Implemented Azure services: Azure AD, Storage, Blob Storage, VM, SQL Database, Functions, ARM, Service Fabric, Service Bus, Cloud Services and IIS.
- Developed Azure Data Factory (ADF) pipelines moving data from on-premises systems to Azure Cosmos DB, Azure SQL Data Warehouse (staging) and Azure ML for scoring, appending scores back to the warehouse.
- Automated the end-to-end model refresh process using Azure Cosmos DB, ADF, Azure ML, Python and PowerShell; delivered deployment tools, staged environments and provisioning scripts.
- Deployed CI/CD with Azure DevOps on Azure Kubernetes Service (AKS) and Docker, configured the AKS cluster autoscaler with Terraform, and implemented monitoring with Azure Monitor and Log Analytics to improve reliability and uptime.

**DevOps Engineer | Usha Martin Technologies, Bengaluru** **Mar 2017 - Jun 2018**

- Implemented CI/CD pipelines using Jenkins, SonarQube, Coverity, GitLab, Terraform and AWS CodePipeline, reducing manual intervention and improving release frequency.
- Managed AWS EC2, S3, RDS, VPC, IAM, Lambda and Route 53; containerized microservices with Docker and Amazon ECS; automated provisioning with Terraform and CloudFormation.
- Monitored with CloudWatch, ELK Stack, Prometheus and New Relic; enforced security through IAM policies, security groups, VPC configuration and regular audits.

**DevOps Engineer | Nixserv IT Solutions Pvt Ltd, Bengaluru** **Sep 2013 - Mar 2017**

- Designed and managed CI/CD pipelines with Jenkins and GitLab CI/CD to automate build, test and deployment.
- Containerized and orchestrated applications with Docker and Kubernetes; implemented Infrastructure as Code (IaC) with Terraform to provision AWS resources.
- Configured monitoring with Prometheus, Grafana and ELK Stack; automated configuration with Ansible; partnered with dev and ops teams on troubleshooting and DevOps best practices.

**Linux Administrator | Bharat Electronics Limited, Bengaluru** **Jul 2012 - Jul 2013**

- Installed and maintained RHEL, CentOS, Ubuntu and Debian servers; managed users, permissions, file systems and access controls.
- Automated administration with Bash/Python and cron; monitored with Nagios; maintained VMware, VirtualBox and KVM; conducted security audits, vulnerability assessments and patch management.

## EDUCATION

**Bachelor of Engineering (B.E.), Electronics and Communication Engineering** **Apr 2011**
Muthayammal Engineering College, Anna University

## CERTIFICATIONS

**AWS Certified Machine Learning - Specialty** | Amazon Web Services (AWS)